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Automating assist identification in football (soccer): a machine learning approach using event and tracking data

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Abstract
Assists—the number of last passes before a goal—have been a predominant metric to evaluate offensive players contribution in association football. This poses two major limitations: first, assist definitions differ across data collection vendors, and second, the focus on passes leading to goals creates a strong dependency on the player taking the shot. We introduce the term potential assists comprising all actions, that prepare a teammate's shot regardless of conversion. Additionally, we use expert-knowledge to derive an objective definition for potential assists and show that it can be detected with a substantial inter-rater reliability (Fleiss' $\kappa = 0.63$, $n = 5$). Using this definition, we utilize video footage to label 500 assists manually and train multiple supervised machine learning models to classify potential assists based on 29 features derived from positional and event data. We apply our automated detection approach on 105 DFB-Pokal matches and demonstrate a scalable method to enhance automatic event detection and improve the comparability of assist statistics for application in player assessment and scouting.

Keywords Machine learning · Football analytics · Soccer · Assist identification · Assist classification · Automatic event detection · Expert labeling

1 Introduction

Association football engages billions of people worldwide [1]. Requiring only a ball and two goals—or creative substitutes—it has grown into a billion-euro industry [2, 3]. From pioneers manually collecting match data in the mid-20th century [4], the field of football analytics now includes a competitive market of data providers offering event and tracking data globally. This data is crucial for scouting, recruitment, performance analysis, and media coverage. y converting match play into quantified events and metrics, football analytics enables more objective evaluations of players, teams, and staff—supporting management, scouting, and coaching decisions.

One use case involves leveraging data to analyze offensive contributions by players including metrics such as goals, expected goals or assists. However, challenges remain due to varying event definitions across vendors and manual collection of events, which introduce subjective interpretations and inconsistencies—even within the same vendor. In this context, vendors refer to commercial data providers offering event and tracking data (e.g., Opta, Wyscout, Sportec Solutions), each using proprietary event definitions and collection procedures. Furthermore, vendors have various assist subtypes differing from each other. For example, Wyscout[1] excludes assists when a decisive dribble

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[1] Wyscout; https://www.hudl.com/en_gb/products/wyscout; Accessed 21 July 2025.

or run occurs before the goal, whereas Opta[2] and FIFA,[3] for instance, base their definitions on the final touch of a teammate, thereby overlooking dribbling actions entirely [5–7]. (Definitions of relevant vendors are listed in Section S1 that can be found in the online supplementary file.) Since event tagging is subjective, operators must decide whether a dribble or run is decisive, adding further variability. Moreover, assists come ahead of scored goals, offensive contribution, though, should comprise potential assists, meaning all actions, that prepare a teammate's shot regardless of conversion.

Automating event collection has been a research topic for over two decades—primarily focusing on events such as shots, passes, and corner kicks—yet potential assists have remained largely unaddressed [8–18]. Similarly, pass quality assessment, which may implicitly reflect assist potential, does not explicitly account for assists, but rather evaluates passes based on their general risk and reward [19–25]. The literature, thus, reveals a gap in an established and objective definition for (potential) assists as well as the automatic detection of such.

This paper fills these gaps due to the following steps:

- After comparing different assist-definitions from literature and data-providers, we suggest an objective definition for assists. The objectivity is backed by an inter-rater-reliability study using video footage.
- We implement a step-by-step feature engineering approach in order to translate positional and event data, i.e. spatio-temporal multi-agent data into a feature vector containing relevant information to classify (potential) assists.
- Based on the above we train multiple supervised machine learning models, trained on hand-labelled assists, to classify arbitrary match situations as assists.
- Lastly, we show potential applications for the resulting classifier and interpret feature importance and interactions.

The following sections review related work (Sect. 2), describe the dataset (Sect. 3) and methodology (Sect. 4, present and visualize results (Sect. 5), discuss findings and limitations, and outline future directions (Sect. 6). Finally, conclusions and key insights are summarized (Sect. 7).

2 Related work

Automatic event detection has been subject of research for over two decades. Tovinkere and Qian [8] introduced an approach to detect events based on tracking data in 2001 by implementing a rule-based approach using domain knowledge for prior modeling of a hierarchical concept and then combining it with tracking information. Also Khaustov and Mozgovoy [9] presented a rule-based algorithm using tracking data for detecting various basic football events like ball possession, passes, and shots on goal. Besides rule-based approaches, Chen et al. [10] applied neural networks on video footage to detect events such as corner-kicks, corner-throws and free-kicks at the box. Likewise, Richly et al. [11] utilized neural networks for pass, kick-off and reception event detection using features calculated on the basis of spatio-temporal data. Gudmundsson and Wolle [12] presented a tool to automatically extract basic events from tracking data, including passes, goals, and fouls. Similarly, Zheng and Kudenko [13] detected events from positional data to generate football commentaries. Machine learning is applied to map known concepts to match states. Richly et al. [14] developed SVM, k-nearest neighbours, and RF models using positional features to detect passes, receptions, and touches. Stein et al. [15] aimed to improve video analysis by detecting both individual and compound events semi-automatically. Fassmeyer et al. [16] used variational autoencoders to learn features from unlabeled data, combined with a large-margin classifier trained on limited labeled samples to detect corner-kicks, crosses, and counter-attacks using event and tracking data, noting that model performance improves with more labeled events. Vidal-Codina et al. [17] applied a decision tree-based algorithm, first computing ball possession, then detecting events based on football-specific assumptions. Recently, Bischofberger et al. [18] introduced an event detection evaluation showing that performance is highly dependent on the dataset's characteristics. Despite advances, assists have not yet been recognized as distinct events in automatic approaches.

Most potential assists originate from passes, therefore research on pass quality evaluation should be considered in assist-related analyses, as it may encompass assessing passes as assists. The development of metrics and statistics for pass evaluation has been in the focus of research for several years. For instance, Anzer and Bauer [19] introduced an expected passes model to assess the difficulty of a pass using spatio-temporal data. Similarly, for example Power et al. [20] measured the expected risk and reward of passes based on tracking data. Chawla et al. [21] employed machine learning to classify passes into three categories: good, ok, and bad. Meanwhile, Szczepański and Mchale [22] extended the completion rate metric to evaluate passing

[2] Opta; https://www.statsperform.com/opta/; Accessed 21 July 2025.

[3] FIFA Training Centre; https://www.fifatrainingcentre.com/en/; Accessed 21 July 2025.

ability by incorporating variables such as the skill of the executing player, the teammate, and random chance. Bransen et al. [23] used event data to compute the expected impact of each pass on the scoreline, analyzing the difference in expected reward before and after the pass. Their work highlighted how passes contribute to creating chances and scoring opportunities. Furthermore, Gyarmati and Stanojevic [24] assessed how passes create more scoring opportunities by analyzing their origin, destination, and location. Finally, the framework of Decroos et al. [25] rated players based on actions that contribute to a team's offensive output, using historical event data.

Assist identification remains unaddressed by automated event detection and pass evaluation studies, showing that the literature review reveals a gap of potential assist identification in prior research.

3 Data

This study uses two key types of match data provided by the German Football Association: event data (e.g. passes, shots, fouls) and tracking data (player, ball, and referee positions). The tracking data have a spatial resolution of 1920×1080 pixels and a temporal resolution of 25 frames per second. In general, data quality and definitions vary by source. This study uses event data from Sportec Solutions[4] and tracking data from Tracab,[5] covering 105 DFB-Pokal matches from the 2023/24 and 2024/25 seasons (up to 31 October 2024), with 2972 shots and 436 goals. The tracking and event data are synchronized providing passes, shots, referee actions, and miscellaneous occurrences covering all remaining events as described by the authors of [19, 26, 27] .

4 Methodology

Our research followed nine linear steps: 1. Defining what constitutes an assist (Sect. 4.1), 2. Selecting 500 shot scenarios (Sect. 4.2), 3. Expert labeling (Sect. 4.3), 4. Data preprocessing (Sect. 4.4) including data preselection (Sect. 4.4.1), domain-specific feature engineering (Sect. 4.4.2), dataset creation (e.g. polynomial features) (Sect. 4.4.3), and feature selection (Sect. 4.4.4). In 5. Four models were selected (Sect. 4.5), and the models were trained and validated (Sect. 4.6) across all scenarios, model types, and feature configurations—resulting in 288 trained models.

The whole process of labeling and scene classification has been supported by domain experts who work as match-analysts or coaches in professional football. The average experience in that area is nine years N = 5, SD = 4.1473 years. All of them hold an UEFA coaching license (A-Level or B-Level).

4.1 Assist definition

While typical assists (e.g. direct passes or crosses) are mutually understood, definitions vary in complex cases. With multiple experts manually labeling data, consistency is crucial. Organizations and vendors like FIFA (with FIFA Football Language), Opta, and Wyscout use broader criteria, while others (e.g. Sportec Solutions) are more restrictive. Detailed assist type definitions for FIFA Football Language, IMPECT,[6] LIGEN,[7] Sportec Solutions, Statsbomb,[8] Opta, and Wyscout can be found in Section S1 located in the online supplementary file. Even if deflected, most definitions include intended passes. Solely, Statsbomb does not refer to the intention of the potential assist giver. However, concepts of intention and ball control remain ambiguous. Some definitions require interpretation. Wyscout, for instance, excludes assists that follow "significant" dribbles or runs, yet the term "significant" is not defined. This study therefore synthesizes multiple definitions into a practical guideline for consistent annotation in close consultation with experts working for the German Football Association. Nevertheless, decisive dribbles and runs represent crucial criteria for defining an assist and often require subjective interpretation, which must be consciously acknowledged.

The following three conditions list, whether an assist is awarded or not.

1. An assist is awarded to a player if the player—regardless of intent—is the last to touch the ball before a teammate's shot on goal. This includes both intended and unintended passes, such as miscontrols, deflections, ricochets, or rebounds off the goalkeeper or the woodwork.
2. An assist is awarded to a player if the player makes an intended distribution (e.g., pass or cross) that successfully reaches the intended teammate and leads to a shot, even if the pass is slightly deflected by an opponent.
3. An assist is *not* awarded to a player if the goal is preceded by a significant individual action, such as a long dribble or run that substantially contributes to the scoring opportunity. Such actions are typically characterized by a notable change in the number of opponents bypassed or the distance reduced between player and

[4] Sportec Solutions AG; https://sportec-solutions.de; Accessed 21 July 2025.

[5] Tracab AB; https://tracab.com; Accessed 21 July 2025.

[6] IMPECT; https://www.impect.com; Accessed 21 July 2025.

[7] LIGEN; https://die-ligen.net; Accessed 21 July 2025.

[8] Hudl Statsbomb; https://www.hudl.com/en_gb/products/statsbomb; Accessed 21 July 2025.

goal. *Example:* Jamal Musiala's dribble preceding his 3:1 goal in FC Bayern München vs. VfL Wolfsburg on 05 February 2023.

Although this definition does not establish concrete boundaries for inclusion or exclusion (e.g., point 3), we show that human experts are capable of classifying scenes with substantial agreement (Fleiss' $\kappa = 0.63$; see [28]).

4.2 Data selection

This research utilizes 500 shot scenarios from the DFB-Pokal, selected according to the criteria outlined in Table 1, ensuring a wide variety as identified through initial analysis of event data and video footage.

For each shot, the preceding pass was identified and the overcome distance and the time difference were calculated. In addition, events were randomly sampled within each category, and the specified number of events was collected randomly. The overcome distance is defined as the Euclidean distance between ball reception and shot location, while the time difference refers to the interval between the receiving and shooting frames.

4.3 Labeling process

The five-phase labeling process began with *Phase 1: Preparation*, in which the authors collected 500 video clips and identified 50 ambigious scenes (i.e. with potentially different perceptions on whether a shot is preceded by an assist or not) through initial labeling. The video scenes included 500 shots on goal, each showing the preceding actions. Experts were asked to label each scenario with '1' for potential assist (indicating that a teammate prepared the shot, regardless of its outcome) or '0' for no potential assist (meaning no teammate's action directly set up the shot). In *Phase 2: Initial Expert Labeling* five domain experts independently labeled these 50 scenes to establish baseline judgments. Across the 50 expert-labeled situations, each individually rated by all five experts, we achieve an inter-rater reliability (IRR) using Fleiss' κ of 0.37 indicating fair agreement as proposed by Landis and Koch [28, p 165]. The IRR measures the level of agreement between two or more independent raters. It assesses whether the labeling process is reliable across different raters and indicates how much confidence can be placed in human-labeled data. Depending on the use case, the IRR can be calculated using different approaches. Having exactly five labelers on all 500 binary scenarios, the Fleiss' κ is used in this study, as it is applicable on more than two raters on a nominal variable [28]. *Phase 3: Expert Exchange* focused on refining the assist definition through collaborative discussion, addressing inter-labeler inconsistencies. In *Phase 4: Full Expert Labeling*, the remaining 450 clips were labeled using the revised criteria, resulting in a substantial IRR of 0.66 (total IRR for all scenarios = 0.63). Finally, *Phase 5: Final Label Creation* aggregated expert inputs into a single label by rounding the average.

Table 1 The eight categories used for the random selection of 500 shot scenarios, along with the number of shots per category

Category	Quantity
Randomly selected	250
Overcome distance in 1st quartile	50
Overcome distance in 2nd and 3rd quartile	25
Overcome distance in 4th quartile	25
Time difference in 4th quartile	50
Successful passes	50
Misdirected passes from own team	25
Misdirected passes from opposing team	25

4.4 Data preprocessing

To prepare the data for model training, relevant events (which might constitute an assist) were selected with a rule-based approach (Sect. 4.4.1), additional features were engineered (Sect. 4.4.2), final datasets were created (Sect. 4.4.3), and feature selection was conducted (Sect. 4.4.4).

4.4.1 Data preselection

The labeled shot scenarios formed the basis for preprocessing. A rule-based system identified the preceding potential assist event, resulting in four assist types across various game situations, as shown in Fig. 1: (1) direct passes (pass event), (2) deflected but successful passes (pass + trajectory check), (3) interjacent teammate touches (miscellaneous), and (4) no valid assist (no assist). The trajectory check is a binary flag marking, that a shot was taken and a pass was played by a teammate beforehand, however, there was another ball touch of an opponent in between. Type 4 shots and edge cases (ambiguous or double-chance shots) were excluded for consistency, leaving 422 shots: 344 with prior successful passes and 78 with unsuccessful prior passes. The final dataset includes features from shot and assist events, pass outcomes, and trajectory analyses.

4.4.2 Feature engineering

Following preprocessing, feature engineering was applied to enhance model performance by transforming existing variables and creating new ones. The applied steps are summarized in Table 2.

Subsequently, the classification target is supported by 29 features, as listed in Table 3, with bolded features exclusively included in the feature sets that incorporate domain

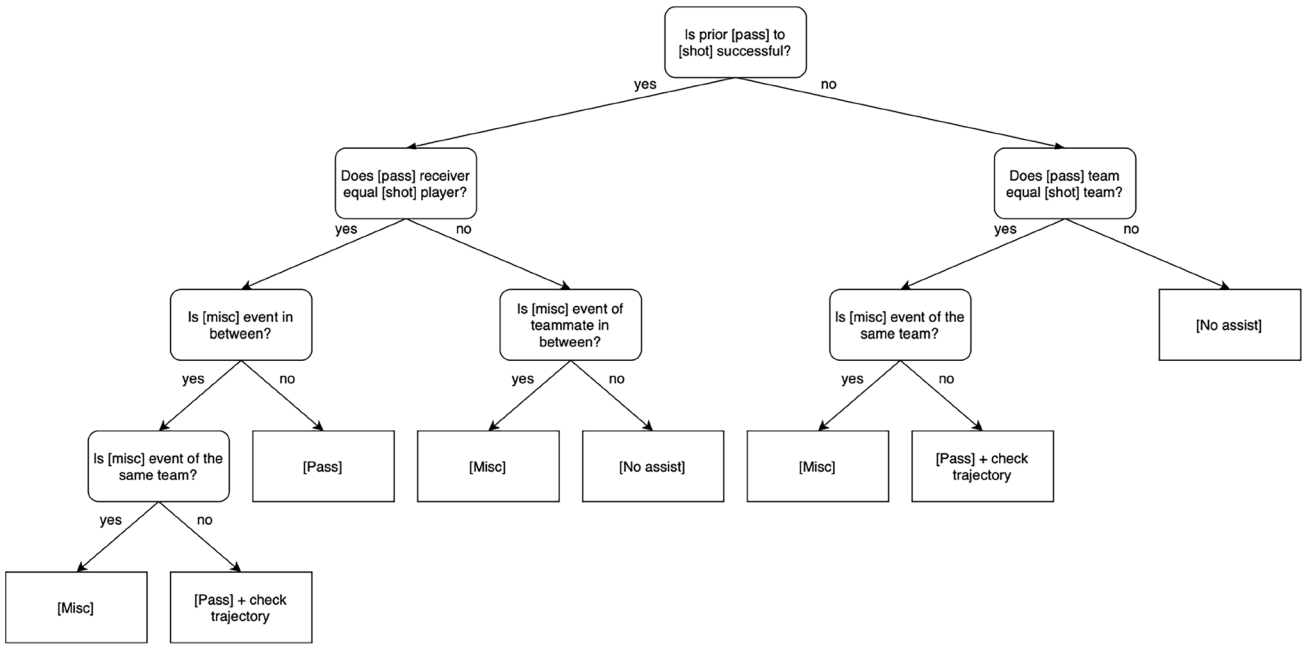


Fig. 1 Decision tree illustrating the rule-based preselection process to identify potential assist events preceding a shot based on event data characteristics, including pass and shot player, team, and event outcome

Table 2 Overview of the nine steps in the feature engineering pipeline

Step	Description
(1) Variable reduction	Remove irrelevant (e.g. kick-off), counterproductive (e.g. assist outcome), and single-value columns
(2) Feature merging and cleanup	Generate interaction-based features (e.g. interval, events in between), then drop obsolete columns
(3) Categorical encoding	Transform player/team IDs to binary variables (e.g. team congruence, player congruence), and encode categorical features
(4) Domain knowledge feature engineering *(optional)*	Construct player-, opponent-, and pass-specific variables (e.g., angle changes, pass trajectory, opponent count) to enhance model interpretability and performance [29, 30]
(5) Correlation filtering	Eliminate highly correlated features to reduce redundancy and multicollinearity
(6) Polynomial feature expansion *(optional)*	Introduce interaction and higher-order terms to capture non-linear effects
(7) Yeo-Johnson transformation	Apply transformation to approximate normal distribution
(8) Feature scaling	Standardize features to a common scale using standardization
(9) Feature selection	Retain relevant features via methods such as Mutual Information, Model Selectors, or Recursive Feature Elimination

knowledge attributes (dataset split explained in the following section). The domain knowledge features comprise manually calculated features, that were not included in the original dataset.

4.4.3 Dataset creation

By selecting events preceding a shot during preselection, it became possible to distinguish successful passes, unsuccessful passes, and other ball touches. This is used to form three clusters based on pass outcomes to evaluate model performance on split datasets individually. This enabled a consideration of successful and unsuccessful prior passes solely and merged, enabling a differentiated model performance evaluation. Additionally, each cluster was used to create a data set using the features of the domain knowledge that are manually engineered (bold in Table 3) and another dataset, that excludes these features. Comparing identical processes on enriched versus non-enriched feature sets and dependent on the prior-pass outcome allows for assessing the impact of manual feature engineering and classifying.

Table 3 Final feature set of the 29 resulting features after preprocessing and feature engineering

Feature	Description	Source type
Assist angle to goal	Angle between assist receivment position and goal	Engineered
Shot angle to goal	Angle between shot position and goal	Engineered
Angle difference	Absolute difference how shot player's angle to goal changed between moment of ball reception and shot	Engineered
Distance to goal from receiving position	Distance from receiver's position to the center of the goal	Engineered
Distance to goal from shot position	Distance from shot location to the center of the goal	Engineered
Change in distance to goal	Difference in distance to the center of the goal before and after the assist	Engineered
Overcome distance	Distance shot player overcomes from pass receipt position to shot position	Engineered
Check trajectory	Boolean indicating if the ball's trajectory should be checked	Engineered
Trajectory angle	Angle between pass trajectory and vector from assist position to ball reception location	Engineered
Frame difference	Number of frames between assist and shot event	Engineered
Number of events in between	Number of other events occurring between assist and shot	Engineered
Shot player equals assist receiver	Boolean indicating if assist was received by the shot taker	Engineered
Shot team equals pass giver	Boolean indicating if both players are from the same team	Engineered
Assist event subtype is none	Indicator for assists without a specific subtype	Original (extracted)
Assist event subtype is corner kick	Indicator for assists originating from a corner kick	Original (extracted)
Assist event subtype is free kick	Indicator for assists from a free kick	Original (extracted)
Assist event subtype is throw in	Indicator for assists originating from a throw-in	Original (extracted)
Assist packing attention	Attention-weighted packing score based on defenders bypassed	Original
Assist packing traditional	Traditional packing score based on players bypassed	Original
Assist XPass	Expected success probability of the assist pass	Original
Assist X tracking of player 1	X-position of the main assist-giving player at the assist moment	Original
Assist Y tracking of player 1	Y-position of the assist-giving player	Original
Assist X tracking of player 2	X-position of secondary assist-contributing player	Original
Assist Y tracking of player 2	Y-position of the secondary contributing player	Original
Shot foot	Foot used for the shot (e.g., left, right)	Original
Shot X tracking of player 1	X-position of the shooter	Original
Shot Y tracking of player 1	Y-position of the shooter	Original
Shot XG	Expected goal value of the shot	Original
Shot player pressure 1	Pressure value on the shooter at the shot moment	Original

Features included exclusively in the domain knowledge feature set are highlighted in bold

4.4.4 Feature selection

The four machine learning models in this study vary in sensitivity to feature set size, correlation, and scaling. While LR and SVM require more preprocessing, tree-based models like RF and XGB are more robust. Feature selection helps address sparsity, redundancy, and overfitting [31, 32]. According to [33], features fall into four relevance categories, and effective selection focuses on both strongly relevant and weakly relevant but non-redundant features. The three main selection methods—filter, wrapper, and embedded—offer trade-offs in speed, complexity, and accuracy [30, 34].

Filter methods quickly rank features using statistical tests (e.g. Mutual Information (MI), correlation) but miss feature interactions. *Wrapper methods* assess subsets by model performance, boosting accuracy but increasing computation and overfitting risk. *Embedded methods*, such as those in RF and XGB, perform selection during training, balancing efficiency and effectiveness. Results can be improved by incorporating expert knowledge [30, 35].

This study applies all three methods alongside expert input. The pipeline includes manual filtering (e.g. removing

correlated features), model-based selectors (RF, XGB, LR), and techniques like MI and Recursive Feature Elimination (RFE). Regularization (L2 for LR, L1/L2 for XGB) supports embedded selection by shrinking less relevant features toward zero.

For each scenario in Fig. 2, five feature selection methods—MI, RFE, and model selectors (RF, XGB, LR)—plus a baseline with no selection, are applied to both the raw feature set and the expanded polynomial set (n = 113). This produces 12 feature sets per scenario. These are then evaluated using all four model types to identify the optimal combination. The full workflow and integration of feature sets with models, using Scenario 1 as an example, are shown in Fig. 3.

4.5 Model selection

We evaluated four machine learning algorithms: LR, SVM, RF, and XGB. LR is simple and efficient, while SVMs excel at handling high-dimensional, non-linear data with kernel functions. As ensemble models, RF and XGB capture complex patterns and support embedded feature selection.

4.6 Training and validation

This research applies a machine learning approach to classify potential assists. Four model types were trained across six scenarios and 12 feature configurations, resulting in 288 models. Each model was evaluated in all combinations of data sets. Missing data was replaced with 0 as other imputing methods (e.g. median imputation) impaired model performance. An initial train/test split of 80/20 was conducted to subsequently perform feature transformation (Yeo-Johnson, standard scaling) and generate synthetic examples in the training set using SMOTE to address class imbalance by generating samples of the minority class overfitting and improves generalization [36]. As the features are engineered and abstract (e.g. spatial and temporal metrics), interpolation produces meaningful variation for a broader training set. That resulted in the following label count: Scenario 1 & 2: class '1'= 216, class '0'= 216 (from 59); Scenario 3 & 4: class '1'= 41 (from 21), class '0'= 41; Scenario 5 & 6: class '1'= 242, class '0'= 242 (from 95). Feature selection was performed on the training data, and applied to the test set. Hyperparameters were optimized using halving random search including cross-validation on the training data, and best hyperparameters were used to assess model performance on the test set.

5 Results

For performance evaluation six metrics were used: F1 score, accuracy, precision, recall, Receiver Operating Characteristic-Area Under the Curve (ROC-AUC), and log loss. The evaluation was conducted across six scenarios, each with two feature sets—one including and one excluding domain-specific features. For each scenario, the four models were trained under 12 configurations comprising polynomial feature expansions and various feature selection techniques.

To ensure clarity given the volume of results, detailed findings are reported only for Scenarios 5 and 6 which

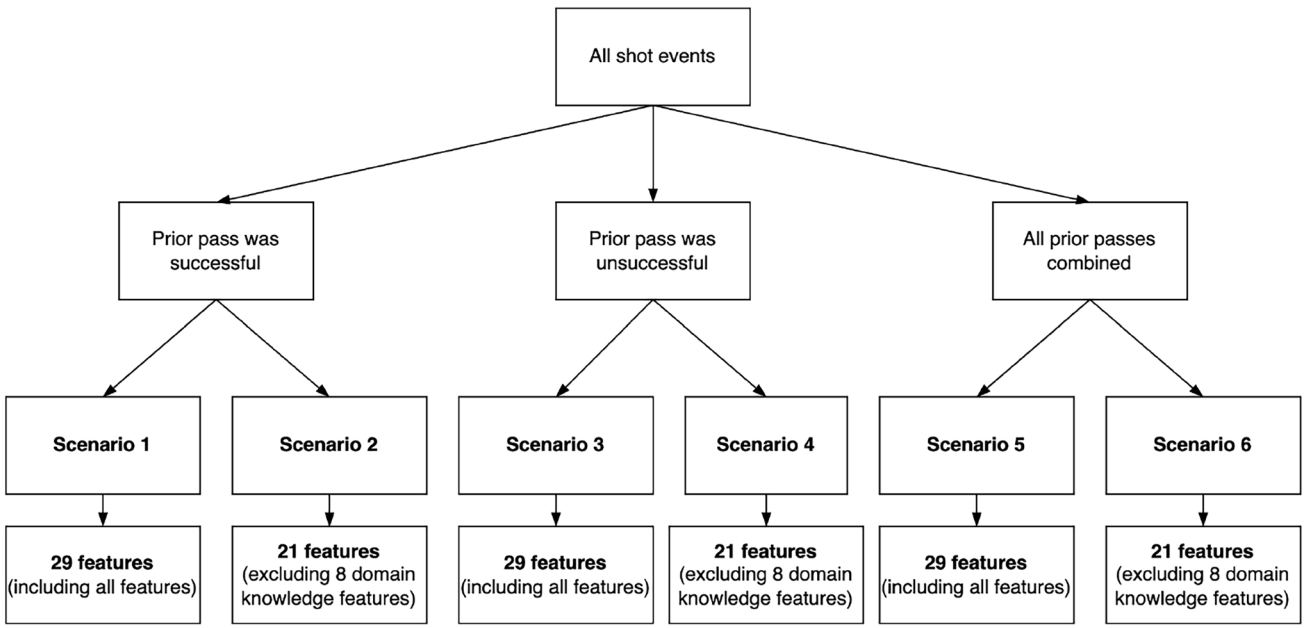


Fig. 2 Overview of the six assist scenarios derived from the original dataset by taking into account the prior pass outcome for individual model training

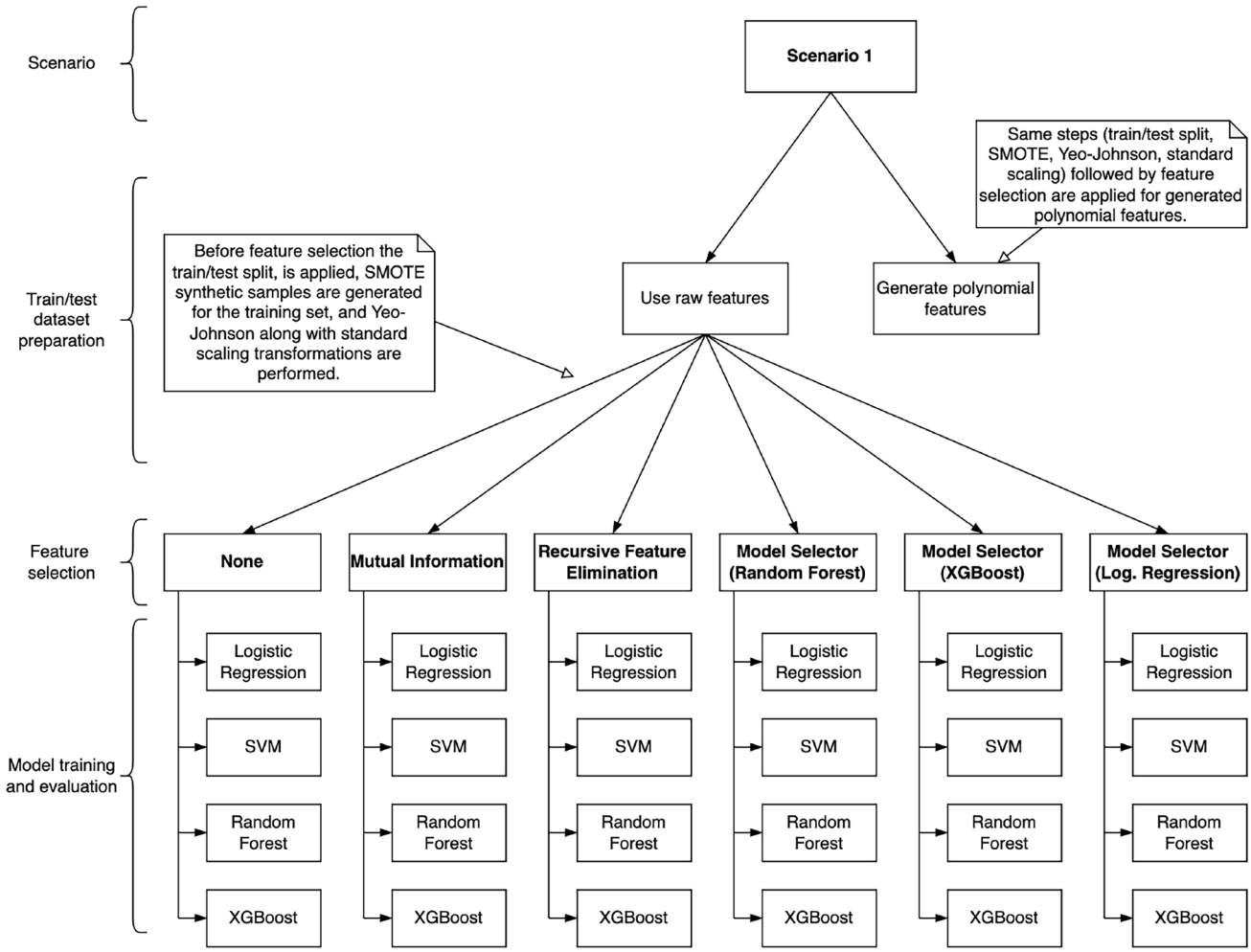


Fig. 3 Presentation of feature selection and subsequent input into model types, exemplary using Scenario 1 with raw features

Table 4 Performance results of the best-performing model for each model type in Scenario 5

Model	Poly[a]	Selection[b]	F1	Precision	Recall	AUC[c]	Log loss
LR	Yes	RF (n = 55)	0.90	0.87	0.92	0.90	0.41
SVM	No	LR (n = 13)	0.91	0.85	**0.98**	0.94	0.32
RF	Yes	LR (n = 100)	**0.94**	**0.92**	0.97	0.94	0.29
XGB	Yes	MI (n = 242)	0.92	0.89	0.96	**0.97**	**0.26**

Values in bold indicate the best-performing model per metric

[a]Poly, automatic generation of polynomial features; [b]Selection, feature selection method and the resulting number of features; [c]AUC, ROC-AUC

encompass all potential assist events. For each scenario, the best-performing configuration per model type is reported; results for Scenarios 1–4 are provided in Section S2 of the online supplementary file.

The F1 score was prioritized during evaluation to address class imbalance. Scenarios 5 and 6 include a total of 422 labeled instances, with 294 classified as potential assists (class '1') and 128 as no potential assist (class '0'). After applying SMOTE for balancing, the training dataset comprised 242 instances of each class, while the test dataset contained 51 instances of class '1' and 33 instances of class '0'. Overall, performance was stable with similar F1 scores across scenarios, with a suitable model identified in each. Ensemble models (RF and XGB) achieved the highest F1 scores in five out of six scenarios and showed the lowest log loss, indicating strong predictive confidence. Differences between the two ensembles were marginal, while LR and SVM generally had higher log loss and lower precision.

In Scenario 5, which clustered all shot scenarios and included domain-knowledge features (see Table 4), RF trained on 100 selected features built on polynomial feature creation—including domain-specific ones—achieved the

highest F1 score (0.94) and precision (0.92), indicating few false positives. RF's performance was followed by XGB (F1 score of 0.92, precision of 0.89), that showed the highest AUC (0.97) and lowest Log Loss (0.26) implying the best distinction and calibration. SVM achieved the highest recall (0.98), permissively capturing nearly all true assists, albeit at the expense of lower precision. This pattern was also observed in Scenarios 1 and 3, where SVM's aggressive recall was offset by increased false positives (precision 0.85).

Excluding domain knowledge features as shown in Table 5 reduced SVM (e.g. F1 score from 0.91 to 0.88, precision from 0.85 to 0.82) and XGB (e.g. L log loss from 0.26 to 0.29) performance. The F1 score was only marginally affected, and model rankings remained largely unchanged. A similar effect was observed between Scenarios 1 and 2, where removing domain knowledge reduced predictive power. SVM's precision and recall dropped notably. As in Scenario 5, RF achieved the highest F1 score (0.94), with a precision of 0.91 and a recall of 0.98. It effectively identified relevant assists (high recall), though lower precision likely reflects the challenge of handling unsuccessful passes, resulting in more risk-tolerant behavior.

Ensemble methods outperformed simpler models like LR and SVM, obtaining the strongest trade-off between precision and recall for the highest F1 scores. Although unsuccessful passes posed challenges for all models, the Random Forest model adapted best, demonstrating effectiveness across event scenarios. The models were more conservative in the challenging Scenarios 3 and 4. Domain knowledge features slightly decreased performance in Scenarios 1 and 2 (metrics have deteriorated by up to 0.03 points) but improved results in Scenarios 3 and 4, emphasizing their value in more complex cases. In particular, 23 of the top 26 models incorporated feature selection, highlighting its role in improving classification, reducing complexity, and improving interpretability.

To assess feature relevance in classifying passes, an interpretability analysis was conducted using SHAP values and Partial Dependence Plots. To ensure consistent feature interpretation, the examination focuses solely on successful passes. As Scenarios 5 and 6 include both successful and unsuccessful preceding passes, the best-performing model from Scenario 2—a Random Forest trained on 21 polynomial features—is selected. The model achieved an F1 score of 0.93, precision of 0.93, and ROC-AUC of 0.87 (Table 9 in Section S2.1 of online supplementary file). Figure 4 displays the 20 most relevant SHAP values for all selected features, ranked by their average impact on predictions. The squared interval between pass and shot (`frame_difference ^2`) had the greatest influence, with lower values increasing the likelihood of an assist. However, some low values corresponded to negative SHAP values, showing no strictly linear relationship. Proximity to the opponent's goal (via `assist_x_tracking_player_1` and its interaction with `shot_x_tracking_player_1`) positively affected assist probability. The `number_of_events` feature showed a minor positive influence, indicating fewer intervening actions increase pass significance. In contrast, tracking values of the shot player, including x- and y-positions, had ambiguous or non-linear impacts. Features like xG, tracking, and packing values showed no single dominant influence, thus SHAP values reveal sensitivity to feature interactions. Also the wide variety of `frame_difference ^2`, or `assist_x_tracking_player_1` highlights the model's sensitivity to context-specific feature interactions.

Furthermore, Partial Dependence Plots in Fig. 5 offer a detailed view of feature relationships and outcomes. As the data is scaled for model training, the x-axis represents transformed values, with x = 0 being the mean. The plots are ordered by SHAP relevance and illustrate key patterns.

Four main dependency trends could be observed:

- *Monotonic increase* Higher feature values consistently increase the likelihood for an assist (e.g. `assist_x_tracking_player_1`, `shot_x_g`).
- *Monotonic decrease*: Increasing values reduce the assist probability (e.g. `shot_y_tracking_player_1`, `number_events_inbetween`).
- *Increase then plateau or dip* Values increase up to the mean, then stabilize or slightly decline (e.g. `shot_x_tracking_player_1`, `assist_x_pass`).
- *Peak at the mean* Values climb up to the mean, followed by a decline (e.g. `shot_y_tracking_player_1`, `assist_packing_attention`).

Table 5 Performance results of the best-performing model for each model type in Scenario 6. Note: Values in bold indicate the best-performing model per metric.

Model	Poly[a]	Selection[b]	F1	Precision	Recall	AUC[c]	Log Loss
LR	No	LR (n = 9)	0.91	0.87	0.94	0.93	0.37
SVM	No	LR (n = 9)	0.88	0.82	0.96	0.94	0.37
RF	Yes	MI (n = 88)	**0.94**	**0.91**	**0.98**	0.94	**0.28**
XGB	Yes	None (n = 113)	0.92	0.88	0.96	**0.96**	0.29

[a]Poly, automatic generation of polynomial features; [b]Selection, feature selection method and the resulting number of features; [c]AUC, ROC-AUC

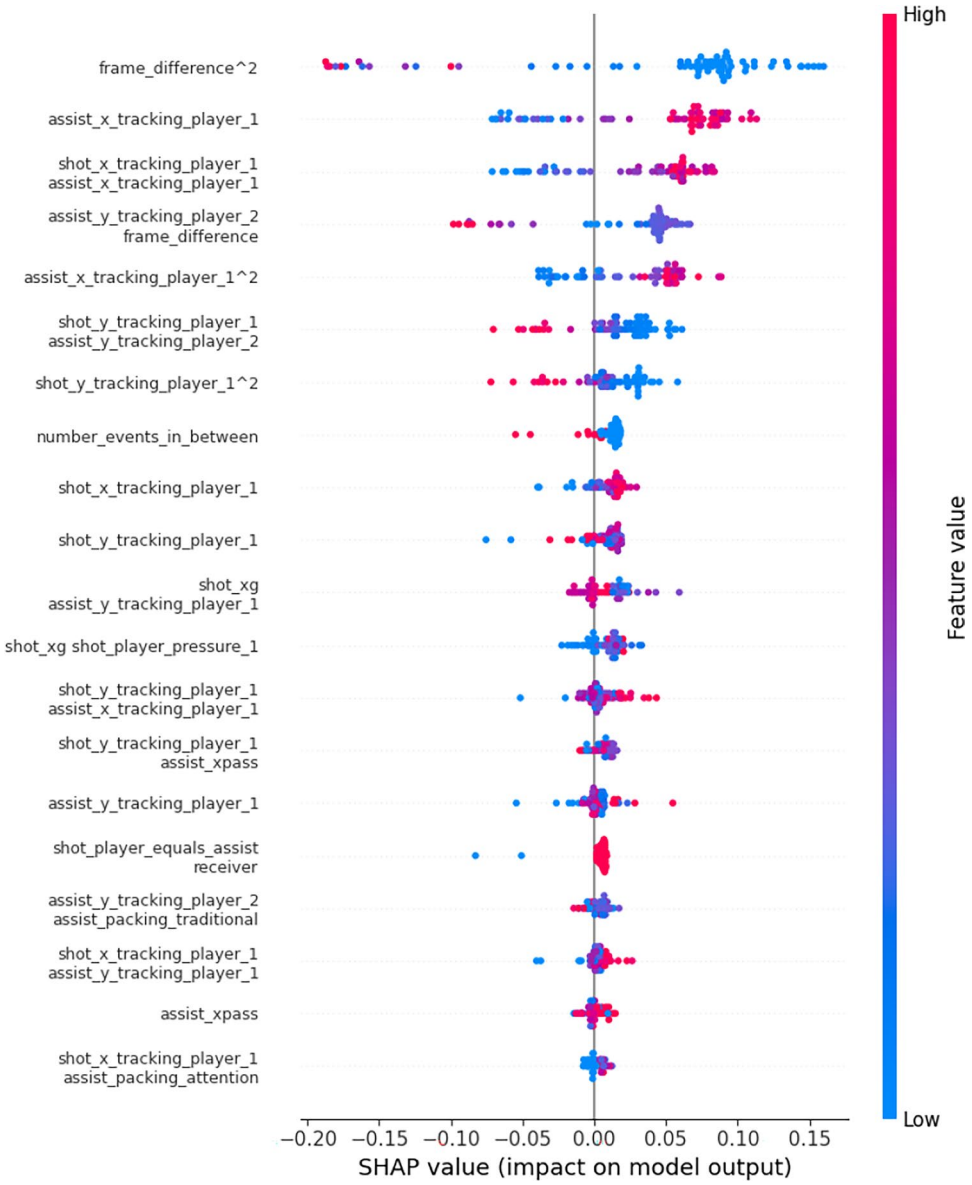


Fig. 4 Top 20 SHAP values for the Random Forest model in Scenario 2

The squared interval (`frame_difference ^2`), for instance, shows higher influence at very low values (< 1), stabilizing around 2.5. The x-position of the passing player (`assist_x_tracking_player_1`) shows a rising impact, with a noticeable increase near the mean before plateauing. A comparable pattern is seen in the interaction between the assist and shot players' x-positions (`shot_x_tracking_player_1 assist_x_tracking_player_1`). Additional non-linear effects appear in features such as `shot_y_tracking_player_1`, `shot_xg assist_y_tracking_player_1`, and `shot_y_tracking_player_1 assist_x_pass`—highlighting the model's ability to capture complex feature interdependencies.

6 Discussion

The presented findings demonstrate successful classification of potential assists using validated assist labels across different scenarios. Ensemble methods like RF and XGB outperformed simpler models (LR and SVM) in capturing complex assist patterns, underlined by RF excelling in five out of six scenarios, particularly in Scenarios 1, 2, 5, and 6. Yet, unsuccessful passes proved challenging for all models, highlighting the importance of feature engineering and expert knowledge and the differentiation in assist formation. The performance drops in Scenarios 3 and 4 suggest the need for further data augmentation and additional features. Enhancing Scenarios 5 and 6, which are biased toward successful passes, could also improve results by expanding the dataset. Moreover, focusing on successful passes solely may

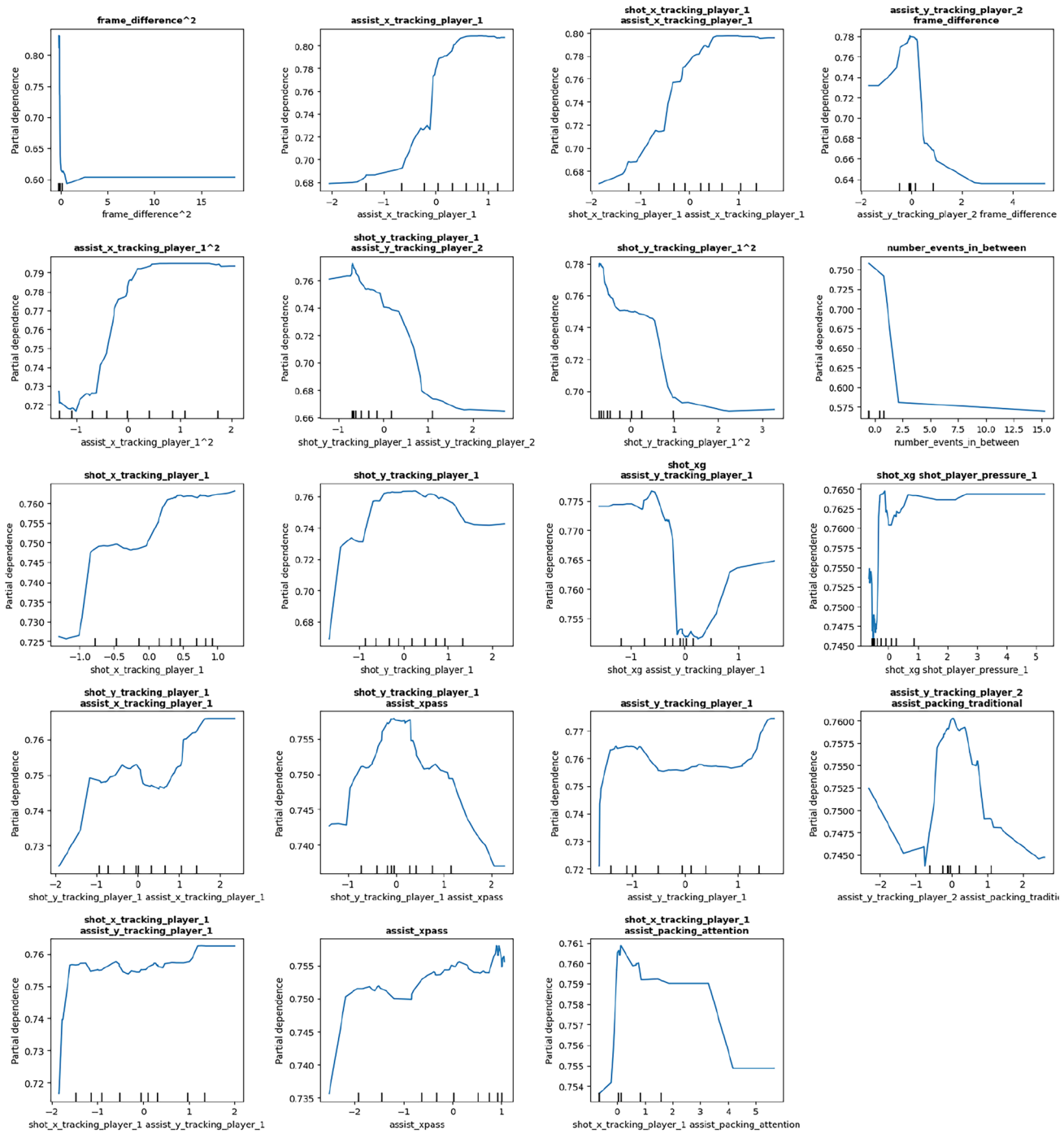


Fig. 5 Partial Dependency Plots for the top 20 SHAP value features (excluding binary feature `shot_player_equals_assist_receiver`) of the Random Forest model in Scenario 2

also boost reliability. Nevertheless, performance above 0.9 must be interpreted considering the IRR of 0.63. Although the model closely approximates the labeled data, the IRR indicates the inherent difficulty and complexity of the classification task also for humans. We introduce an assist definition developed with experts from the German Football Association considering the various characteristics of relevant vendors. Nonetheless, this definition is not without limitations—highlighting the need for a thorough, dedicated study on assist definitions.

This study merges growing areas of football analytics including automatic event detection and pass quality assessment. Previous research has applied machine learning to extract events from tracking data [17, 18] or used

rule-based methods [9]. Others have automatically identified basic events like passes or ball touches [11, 14], but typically focus on simple events without addressing more complex concepts like assists. Detecting assists requires not only identifying a pass, but also evaluating the context that qualifies it as a goal-creating action. Recent studies on pass quality assess factors like complexity, risk-reward, or expected impact on scoring [19, 20, 37]. For example, [37] analyze player involvement in chance creation but do not specifically identify potential assists. This work combines automatic event detection with pass quality evaluation to identify assist events, including successful and failed passes, and other ball actions. It offers more detailed player evaluation and cross-competition comparability. To our knowledge, this is the first study focused on the automatic detection of assists, making it a novel contribution to the field.

A framework for automatic assist identification is presented in this research, combining expert knowledge and a classification approach to enhance the analysis of assist challenges. A common, practical definition of an assist was developed by integrating existing definitions, highlighting the challenges of standardizing assist criteria across the football world. By subdividing potential assist scenarios, the labeling process incorporated various assist categories, preserving randomness, and merging expert opinions into a consistent dataset to illustrate task complexity. Four models, tested across six scenarios, optimized performance through feature transformation, selection, hyperparameter tuning, and cross-validation. Expert-based feature engineering emphasized practical relevance, while SHAP values and Partial Dependency Plots offered insights into model behavior and feature interactions. In Scenario 6, the RF model achieved an F1 score of 0.94, balancing precision and recall, while it scored 0.93 and 0.89 in Scenarios 1 and 3, highlighting the benefits of separating pass outcomes. To identify strengths and weaknesses the performance was assessed using F1 scores, precision, recall, ROC-AUC, and log loss.

The final dataset comprised 422 randomly selected shot scenarios. This work establishes a solid foundation and encourages further research, supported by promising initial results. To address class imbalance in the training set, SMOTE was applied, which introduces the potential limitation of generating synthetic examples that may not fully reflect real-world football situations. However, this risk is mitigated by the use of abstract, engineered features and validation on authentic examples. Nonetheless, the application of the model to entirely new scenarios has to be obeserved.

Given the relatively small sample size, the high F1 scores may indicate that the features are both informative and well-engineered. However, this also raises the risk of overfitting, and the model's ability to generalize may be limited when applied to broader datasets. To address this, and to capture a wide variety of potential assist situations, the scenario categories listed in Table 1 were deliberately chosen to ensure broader representativeness and reduce the risk of bias. More samples could address the imbalance, partially mitigated by synthetic examples.

Performance in Scenarios 3 and 4 may be limited by the smaller sample size, that was also biased due to the selection process explained in Table 1 while Scenarios 5 and 6 faced imbalance in the ratio of successful to unsuccessful passes. Missing or noisy data reduced training quality but could improve generalizability. Time differences between the assist and shot were calculated using frames, though defaults were assumed for miscellaneous events, potentially impairing accuracy. Event data categorized four types (pass, shot, miscellaneous, referee), but differing event types from various providers might complicate assist mapping. The rule-based approach works well for patterns (e.g. successful pass to shooter) but struggles with more complex events, such as those involving multiple actions (e.g. corner kicks).

Our findings demonstrate a well-performing automatic assist detection using event and tracking data, offering a foundation for improved player performance evaluation. Automated assist identification will be essential for future assessments grounded on automated event data collection, helping identify underrated players and influencing scouting, contracts, and transfers. SHAP values provided interpretability, revealing key features for assist detection, while Partial Dependency Plots highlighted non-trivial patterns for detailed event analysis.

This research lays a solid foundation and shows, that automatic identification of potential assists is possible, however, further validation and refinement of the model are required. Experts could retrospectively label additional matches, and their feedback—combined with insights from SHAP values and partial dependence plots—may further improve performance. Additionally, our potential assist framework could complement metrics such as expected assists (xA) by using the identification of potential assists as the basis for awarding xA values to players. Expanding the dataset, particularly for unsuccessful passes, is crucial to improve balance and address challenges with these events. Additionally, the rule-based method for selecting the potential assist event could be improved by applying machine learning methods. For practical application, assists resulting from set-piece creation (e.g. direct free-kicks, penalties) should also be included. Finally, to advance the system, it could be adapted to use only tracking data, potentially leveraging existing frameworks or deep learning for assist detection.

7 Conclusion

This study presented a framework for automatic assist detection in football based on combined event and tracking data. The work contributes to the broader field of automatic event detection and supports data-driven player performance evaluation.

The results showed that ensemble models, particularly Random Forest and XGBoost, consistently outperformed simpler baselines and achieved reliable classification of potential assists across the defined scenarios (best F1 = 0.94).

Incorporating domain knowledge, targeted feature engineering and selection, as well as hyperparameter tuning proved essential for achieving robust performance.

The findings advance football analytics by demonstrating that automatic assist detection can enhance the comparability of assist-related metrics across competitions and data vendors, supporting more objective assessment of offensive contributions. However, the approach remains constrained by the quality and consistency of human-labeled training data (Fleiss' $\kappa = 0.63$) and by the limited number of annotated matches. Future work should extend validation across larger datasets and evaluate real-time deployment scenarios.

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Data availability The data is available upon reasonable request.

Declarations

Conflict of interest The authors have no Conflict of interest to declare that are relevant to the content of this article.

Ethical approval This study used secondary match data provided by the German Football Association (DFB). No human participants or personally identifiable information were involved; therefore, formal ethical approval was not required.

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